MH1200 Linear Algebra I.

Problem set #8.

SOLUTIONS

Core problems:

Solution to Problem 1: test your knowledge on the axioms of subspace

In the following, when showing that a subset is not a subspace, we do it by showing either (i) it doesn't contain the zero vector, or (ii) it isn't closed under addition, or (iii) it isn't closed under scalar multiplication.

To show that a subset is a subspace, we should show that it satisfies the 3 conditions for a being subspace, namely, contains the zero-vector and is closed under addition and scalar multiplication. But actually it can usually be done more easily by using some theorems from the lectures. So in the following the solutions are done using those theorems. As an exercise you should also try to show explicitly that those subsets below which are subspaces do satisfy the requirements of containing the zero-vector and being closed under addition and scalar multiplication.

1. $\{ (0,0,0) \}$.

   This is a subspace: it is called the zero subspace. Note that it is (0) non-empty (i) closed under vector addition because $(0,0,0)+(0,0,0) = (0,0,0)$ (ii) closed under scalar multiplication because for any $r \in \mathbb{R}$, $r(0,0,0) = (0,0,0)$.

2. $\{ (1,1,1) \}$.

   Not a subspace: it doesn't contain the zero vector. We showed in lectures that every subspace must contain the zero vector.

3. $\{ (0,0,0),\ (1,1,1) \}$.

   Not a subspace: it's not closed under scalar multiplication because it contains $(1,1,1)$ but not $\frac{1}{2}(1,1,1) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.

4. $\{ (0,0,c); c \text{ is an integer} \}$.

   Not a subspace: this is not closed under multiplication because for example it contains $(0,0,1)$ but not $\frac{1}{2}(0,0,1) = (0,0,\frac{1}{2})$. (This third coordinate cannot be a non-trivial fraction like this. The subset only has integers in that position.)

5. $\{ (0,0,c); c \text{ is a non-negative real number} \}$.

   Not a subspace: it contains $(0,0,1)$ but not $-1(0,0,1) = (0,0,-1)$.

6. $\{ (0,0,c); c \text{ is a real number} \}$.

   This is a Subspace: it is $\text{span}\{(0,0,1)\}$ since $(0,0,c) = c(0,0,1)$. (By a theorem from the lectures every span of a collection of vectors is a subspace.)

   Alternatively, if you are still understanding what span is, we can check the axioms directly. The set is obviously non-empty. Furthermore it is closed under vector addition because $(0,0,c)+(0,0,d) = (0,0,c+d)$ which is again in the subset. Finally it is closed under scalar multiplication because $r(0,0,c) = (0,0,rc)$.

7. $\{ (1, 1, c); c$ is a real number $\}$.

   Not a subspace: it doesn't contain $(0, 0, 0)$.

8. $\{ (a, b, c); a,\ b,\ c$ are real numbers and $a \geq b \geq c \}$.

   Not a subspace: it contains $(1, 0, 0)$ because $1 \geq 0 \geq 0$ but not $-1(1, 0, 0) = (-1, 0, 0)$ because -1 is not ≥ 0.

9. $\{ (a, b, c); a,\ b,\ c$ are real numbers and $4a = 3b \}$.

   This is indeed a subspace: it is in fact the solution set to the homogeneous linear equation $4a - 3b = 0$. (By a theorem from the lectures the solution set of a homogeneous linear system is always a subspace.)

10. $\{ (a, b, b); a,\ b$ are real numbers $\}$.

    Subspace: it is $\text{span}\{ (1, 0, 0),\ (0, 1, 1)\}$ since $(a, b, b) = a(1, 0, 0) + b(0, 1, 1)$. Thus it is a subspace.

    Alternatively we can check the 3 axioms directly. It is non-empty. And it is closed under vector addition because $(a_1, b_1, b_1) + (a_2, b_2, b_2) = (a_1 + a_2, b_1 + b_2, b_1 + b_2)$ which is again of the form defining the subset.

11. $\{ (a, b, ab); a,\ b,\ c$ are real numbers $\}$.

    Not a subspace: it contains $(1, 0, 0)$ and $(0, 1, 0)$ but not their sum $(1, 0, 0)+(0, 1, 0) = (1, 1, 0)$.

12. $\{ (a^2, b^2, c^2); a,\ b,\ c$ are real numbers $\}$.

    Not a subspace: it contains $(1, 0, 0)$ but not $-1(1, 0, 0) = (-1, 0, 0)$.

13. $\{ (a^3, b^3, c^3); a,\ b,\ c$ are real numbers $\}$.

    Subspace: it is the whole of $\mathbb{R}^3$ since every $(x, y, z) \in \mathbb{R}^3$ can be written as $(x, y, z) = (a^3, b^3, c^3)$ where a, b, c are the cubed roots of x, y, z respectively.

14. $\{ (x, y, z); x, y, z \in \mathbb{R} \quad \text{and} \quad x + y - 2z = 0\}$.

    Subspace: it is the solution set of a homogeneous linear equation.

15. $\{ (x - 1, y, z); x, y, z \in \mathbb{R} \quad \text{and} \quad x + y - 2z = 0\}$.

    Not a subspace: We can check that the zero vector is not in this subset. To get the zero vector $(0, 0, 0)$ the parameters need to be $x = 1$, $y = 0$ and $z = 0$. But these choices of parameters do not satisfy the equation $x + y - 2z = 0$.

16. $\{ (x, y - 1, z); x, y, z \in \mathbb{R} \quad \text{and} \quad x + y - 2z = 1\}$.

    Subspace: Setting $y' = y - 1$ we see that it is the same subset as $\{ (x, y', z); x, y', x \in \mathbb{R} \quad \text{and} \quad x+y'+1-2z = 1\}$. The condition $x+y'+1-2z = 1$ is the same as the homogeneous linear equation $x + y' - 2z = 0$, so this subset is the solution set of a homogeneous linear equation and is therefore a subspace by a theorem from the lectures.

17. $\{\,(x_1, x_2, x_3, x_4); x_1, x_2, x_3, x_4 \in \mathbb{R} \quad \text{and} \quad x_1 = x_2 + x_3 = x_4 - x_3 + 2x_1\}$.

Subspace: The condition $x_1 = x_2 + x_3 = x_4 - x_3 + 2x_1$ is the same as the pair of equations $x_1 = x_2 + x_3$ and $x_2 + x_3 = x_4 - x_3 + 2x_1$. These can be rewritten as a homogeneous linear system:

$$\begin{aligned} x_1 - x_2 - x_3 &= 0 \\ x_2 + 2x_3 - x_4 - 2x_1 &= 0 \end{aligned}$$

So the subset is the solution set of a homogeneous linear system and is therefore a subspace.

□

Problem 2: test knowledge on intersection of sets and subspace

Let U, V, W be three planes in $\mathbb{R}^3$ where

$$U = \{\,(x, y, z); 2x - y + 3z = 0\,\},\ V = \{\,(x, y, z); 3x + 2y - z = 0\,\},$$
$$W = \{\,(x, y, z); x - 3y - 2z = 1\,\}.$$

1. Determine which of U, V, W contain the origin.
2. Give parameterizations for the sets $U \cap V$ and $V \cap W$. (In other words: find a parameterization for the general solution of the corresponding linear system which describes the intersection).
3. Is $U \cap V$ a subspace of $\mathbb{R}^3$? Is $V \cap W$ a subspace of $\mathbb{R}^3$? Justify your answers.

Solution to Part (1).

U contains the origin since $2x - y + 3z = 0$ is satisfied by $(x, y, z) = (0, 0, 0)$.
V contains the origin since $3x + 2y - z = 0$ is satisfied by $(x, y, z) = (0, 0, 0)$.
W does not contain the origin since $x - 3y - 2z = 1$ is not satisfied by $(x, y, z) = (0, 0, 0)$.

Solution to Part (2).

The intersection $U \cap V$ is the set of points which lie in both planes simultaneously. In other words: it is the solution set of the following linear system:

$$\begin{aligned} 2x - y + 3z &= 0 \\ 3x + 2y - z &= 0 \end{aligned}$$

This is solved in the usual way by writing down the augmented matrix of the system, applying Gaussian Elimination to reduce it to row-echelon form, and then find the general solution by back-substitution. The result is:

$$U \cap V = \{\,(t, -\tfrac{11}{5}t, -\tfrac{7}{5}t); t \in \mathbb{R}\}.$$

Next: $V \cap W$ is the solution set of the following linear system:

$$\begin{aligned} 3x + 2y - z &= 0 \\ x - 3y - 2z &= 1 \end{aligned}$$

The result for the general solution in this case is

$$V \cap W = \{\,(-\tfrac{1}{5} - \tfrac{7}{5}t,\ t,\ -\tfrac{3}{5} - \tfrac{11}{5}t); t \in \mathbb{R}\}.$$

Solution to Part (3).

In part 2 we saw that $U \cap V$ is the solution set of a homogeneous linear system. So it is a subspace acording to a theorem from the lectures.

In part 2 we also saw that $V \cap W$ is the solution of an non-homogeneous linear system. So, it cannot be a subspace, because the origin will not be a solution to the system.

Another, simpler, argument for this is to recall from part 1 that W does not contain the zero vector; hence $V \cap W$ doesn't contain the zero vector and therefore cannot be a subspace.

$\square$

Problem 3: proof + simple qn

Let V be a subspace of $\mathbb{R}^3$. Define $W = \{(x, z); (x, y, z) \in V\}$. (In words, W is the subset of $\mathbb{R}^2$ consisting of pairs of numbers with the property that the pair is the first coordinate and the third coordinate of some vector in the subspace V.)

(i) Show that W is a subspace of $\mathbb{R}^2$.

(ii) What is W if $V = \{(x, y, 0) : x, y \in \mathbb{R}\}$?

Solution to (i).

To show that W is a subspace we'll check the three axioms directly in turn.

- W is nonempty: $(0, 0) \in W$ since $(0, 0, 0) \in V$ (we know V is a subspace and every subspace contains the zero vector).
- W is closed under addition: Let $(x_1, z_1), (x_2, z_2) \in W$. By the definition of W there are $y_1, y_2 \in \mathbb{R}$ with $(x_1, y_1, z_1) \in V$ and $(x_2, y_2, z_2) \in V$. Since V is a subspace of $\mathbb{R}^3$, we have $(x_1, y_1, z_1) + (x_2, y_2, z_2) = (x_1 + x_2, y_1 + y_2, z_1 + z_2) \in V$. By the definition of W this implies $(x_1 + x_2, z_1 + z_2) \in W$ and hence $(x_1, z_1) + (x_2, z_2) \in W$.
- W is closed under scalar multiplication: Let $(x, z) \in W$ and $\lambda \in \mathbb{R}$. By the definition of W there is $y \in \mathbb{R}$ such that $(x, y, z) \in V$. Since V is a subspace of $\mathbb{R}^3$, we conclude $\lambda(x, y, z) = (\lambda x, \lambda y, \lambda z) \in V$. By the definition of W, this implies $(\lambda x, \lambda z) \in W$. Hence $\lambda(x, z) \in W$.

Solution to (ii).

$$
\begin{aligned}
W &= \{(x, z) : \text{ there is } y \in \mathbb{R} \text{ with } (x, y, z) \in V\} \\
&= \{(x, z) : x \in \mathbb{R}, z = 0\} \\
&= \{(x, 0) : x \in \mathbb{R}\}. \text{ This is precisely the } x\text{-axis in } \mathbb{R}^2.
\end{aligned}
$$

$\square$

test your understanding on the axioms of subspace

Problem 4: (Assembled from different versions of a Quiz in 2019.)

Consider the following subsets of $\mathbb{R}^3$. In each case, state whether the subset is or is not a subspace of $\mathbb{R}^3$. In each case justify your answer.

(i) $S_1 = \{(a,b,c) \in \mathbb{R}^3; a+b=0, b+c+1=0\}$.

(ii) $S_2 = \left\{ (a,b,c) \in \mathbb{R}^3; \begin{vmatrix} 1 & 2 & 3 & 4 \\ a & 0 & b & 0 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 14 \end{vmatrix} = 0 \right\}$.

(iii) $S_3 = \left\{ (a,b,c) \in \mathbb{R}^3; \begin{vmatrix} 1 & 1 & 1 \\ a^2 & b & c \\ 2 & 1 & 1 \end{vmatrix} = 0 \right\}$.

(iv) $S_4 = \{(a,b,c) \in \mathbb{R}^3; a+b=0, b+c=0\}$.

(v) $S_5 = \left\{ (a,b,c) \in \mathbb{R}^3; \det\left(\begin{bmatrix} 1 & a^2 & 1 \\ 1 & b^2 & 1 \\ 1 & c^2 & 1 \end{bmatrix} \right) = 0 \right\}$.

(vi) $S_6 = \left\{ (a,b,c) \in \mathbb{R}^3 : \begin{vmatrix} 1 & 2 & 3 & 4 \\ a^2 & 0 & b^2 & 0 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{vmatrix} = 0 \right\}$.

Solution to (i).

This subset is not a subspace. Note that it does not contain the zero vector $a=b=c=0$ because $0+0+1 \neq 0$. But we know that every subspace contains the zero vector, so this can't be a subspace.

Solution to (ii).

This subset is a subspace. To see this: calculate the determinant in the given formula using a cofactor expansion across the second row to obtain

$$a\left(- \begin{vmatrix} 2 & 3 & 4 \\ 6 & 7 & 8 \\ 10 & 11 & 12 \end{vmatrix} \right) + b\left(- \begin{vmatrix} 1 & 2 & 4 \\ 5 & 6 & 8 \\ 9 & 10 & 12 \end{vmatrix} \right) = 0.$$

We don't need to calculate these two determinants. We can identify straight away that this set is the set of solutions of a homogeneous linear system, so by a standard theorem, this is a subspace.

Solution to (iii).

This subset is a subspace. To see this: calculate the determinant in the given formula using a cofactor expansion across the second row to obtain

$$a^2\left(- \begin{vmatrix} 1 & 1 \\ 1 & 1 \end{vmatrix} \right) + b\left(\begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} \right) + c\left(- \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} \right) = a^2(0) - b + c = 0.$$

We don't need to calculate these two determinants. We can identify straight away that this set is the set of solutions of a homogeneous linear system, so by a standard theorem, this is a subspace.

Solution to (iv).

This subset is a subspace because we can recognise immediately it is the set of solutions of a system of homogeneous equations.

Solution to (v).

This subset is a subspace. To see this: note that because two of the columns of this determinant are equal, this determinant is always zero. So the given equation is always satisfied, no matter what a, b and c are. Thus the set is actually $\mathbb{R}^3$. By definition this is a subspace of $\mathbb{R}^3$.

Solution to (vi).

Again this set is a subspace. The reason is that we can compute the determinant to be zero. Below we do row operations: $R_3 \to R_3 - R_1$, $R_4 \to R_4 - R_1$ then factor out the 2 multiplying row 4. The result is matrix with two identical rows which will have determinant zero.

$$\begin{vmatrix} 1 & 2 & 3 & 4 \\ a^2 & 0 & b^2 & 0 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 & 4 \\ a^2 & 0 & b^2 & 0 \\ 4 & 4 & 4 & 4 \\ 8 & 8 & 8 & 8 \end{vmatrix} = 2 \begin{vmatrix} 1 & 2 & 3 & 4 \\ a^2 & 0 & b^2 & 0 \\ 4 & 4 & 4 & 4 \\ 4 & 4 & 4 & 4 \end{vmatrix} = 0.$$

This equation is always satisfied, no matter what a and b are.

$\square$

Problem 5: (Appeared on the final exam 2018.) test your knowledge on the axioms of subspace

(a) Write down the formula for the trace of the matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$.

Now consider the following subsets of $\mathbb{R}^4$. In each case either briefly prove the given set is a subspace of $\mathbb{R}^4$, or briefly prove it is not a subspace.

(b) $S_1 = \left\{(a,b,c,d) \in \mathbb{R}^4 : \operatorname{tr}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = 0\right\}$. yes

(c) $S_2 = \left\{(a,b,c,d) \in \mathbb{R}^4 : \operatorname{tr}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) \geq 0\right\}$. no

(d) $S_3 = \left\{(a,b,c,d) \in \mathbb{R}^4 : \operatorname{tr}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}^2\right) = 0\right\}$. no

(e) $S_4 = \left\{(a,b,c,d) \in \mathbb{R}^4 : \det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}^T\right) \geq 0\right\}$. yes

The simplest approach to this problem was to evaluate the given matrix expressions directly, then to study the resulting sets uses the techniques we know for checking the subspace axioms.

Solution to (b).

In this case the set is defined by the expression

$$S_1 = \left\{(a,b,c,d) \in \mathbb{R}^4 : a + d = 0\right\}$$

This is obviously the set of solutions of a homogeneous linear system, so by the standard theorem it is a subspace.

Solution to (c).

In this case the set is defined by the expression

$$S_2 = \left\{(a,b,c,d) \in \mathbb{R}^4 : a + d \geq 0\right\}$$

This is not a subspace because it is not closed under scalar multiplication. For example $(1,0,0,0) \in S_2$ because $1 + 0 \geq 0$ but $-1(1,0,0,0) = (-1,0,0,0)$ does not lie in S_2.

Solution to (d).

In this case we need the expression

$$\operatorname{tr}\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = \operatorname{tr}\left(\begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}\right) = a^2 + 2bc + d^2.$$

Thus the set becomes

not a linear eqn

not homogeneous

$$S_3 = \left\{(a,b,c,d) \in \mathbb{R}^4 : a^2 + 2bc + d^2 = 0\right\}$$

This set is not a subspace because, say $(1,-1,1,1)$ and $(1,1,-1,1)$ lie in S_3, but $(1,-1,1,1) - (1,1,-1,1) = (0,-2,2,0)$ does not.

Solution to (e).

This part was a bit of a trick question. If we calculate the given matrix expression in this case, the first step is to use the fact that determinant respects multiplication. We get:

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix}^T\right) = \det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right)\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}^T\right)$$

Because determinant is not changed by taking the transpose, the inequality defining the set becomes:

$$\det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right)^2 \geq 0.$$

The square of a real is ALWAYS ≥ 0 so this condition is ALWAYS satisfied. Thus

$$S_4 = \mathbb{R}^4.$$

Yes by definition this is a subspace of $\mathbb{R}^4$.

$\square$

Problem 6 proof + give example **(From final exam 2019.)**

(a) Consider the following subset of $\mathbb{R}^3$

$$S = \left\{(x_1, x_2, x_3) \in \mathbb{R}^3,\ x_1 x_2 x_3 = 0\right\}.$$

Either prove that it is a subspace of $\mathbb{R}^3$ or prove that it is not.

(b) Give an example of a non-empty subset U of $\mathbb{R}^2$ simultaneously satisfying the two properties that

- U is closed under vector addition
- whenever $\mathbf{u} \in U$ then $(-1)\mathbf{u} \in U$ as well

but which is not a subspace of $\mathbb{R}^2$.

Solution to (a).

This is not a subspace. For example $(1,1,0) \in S$ and $(1,0,1) \in S$ but

$$(1,1,0) + (1,0,1) = (2,1,1)$$

does not lie in S. Thus this set is not closed under vector addition.

Solution to (b).

An example might be:

$$U = \left\{(m,n) \in \mathbb{R}^2, m, n \in \mathbb{Z}\right\}.$$

Namely the vectors in $\mathbb{R}^2$ with **integer** entries. It clearly satisfies the two stated properties, but is not a subspace because it isn't closed under scalar multiplication. For example: $\frac{1}{2}(1,1) \notin U$ even though $(1,1) \in U$.

□

Problem 7: (2.5 marks.)

proof

Let:

- n be a positive integer.
- Let $\mathbf{M}$ be an $n \times n$ matrix.

If $\vec{v} \in \mathbb{R}^n$ then let $\mathbf{M}_{\vec{v}}$ denote the matrix that results when you replace the last column of $\mathbf{M}$ with the vector $\vec{v}$.

Directly using the axioms of a subspace, carefully prove that the set

$$S = \{\vec{v} \in \mathbb{R}^n ; \det(\mathbf{M}_{\vec{v}}) = 0\} \subset \mathbb{R}^n$$

is a **subspace** of $\mathbf{R}^n$. You may use standard properties of determinants discussed in the course.

Solution

We'll check the three axioms of a subspace in turn for S.

(Axiom 1: The set contains the zero vector.)

The determinant of a matrix containing a zero column is zero. Thus $\det(\mathbf{M}_{\vec{0}}) = 0$. Thus $\vec{0} \in S$.

(Axiom 2: The set is closed under vector addition.)

Let $\vec{v} \in \mathbb{R}^n$ and $\vec{w} \in \mathbb{R}^n$ be vectors in S. We must show that $\vec{v} + \vec{w} \in S$.

Because $\vec{v}, \vec{w} \in S$ we know that

$$\det(\mathbf{M}_{\vec{v}}) = \det(\mathbf{M}_{\vec{w}}) = 0.$$

Now because of the standard property that determinants are linear functions of specific columns, we know that

$$\det(\mathbf{M}_{\vec{v}+\vec{w}}) = \det(\mathbf{M}_{\vec{v}}) + \det(\mathbf{M}_{\vec{w}}) = 0 + 0 = 0.$$

Thus $\vec{v} + \vec{w} \in S$. Thus S is closed under vector addition.

(Axiom 3: The set is closed under scalar multiplication.)

Let $\vec{v} \in S$ and let $c \in \mathbb{R}$. Because $\vec{v} \in S$ we know that

$$\det(\mathbf{M}_{\vec{v}}) = 0.$$

Because determinants are linear functions of specific columns we know that

$$\det(\mathbf{M}_{c\vec{v}}) = c\det(\mathbf{M}_{\vec{v}}) = c.0 = 0.$$

Thus $c\vec{v} \in S$. Thus S is closed under scalar multiplication.

In conclusion: S satisfies the 3 axioms of a subspace, hence it is a subspace. $\square$

Comment

Another approach would be to observe that the condition $\det(\mathbf{M}_{\vec{v}}) = 0$ becomes a single homogeneous linear equation and then check the three axioms hold in turn for the set of solutions of such an equation. It is not enough just to observe it is homogeneous because the problem asks for a careful proof from the axioms.

proof

Problem 8:

In this problem we'll interpret $\mathbb{R}^n$ to be the set of column vectors with n entries. In other words, instead of writing the list horizontally

$$(u_1, \ldots, u_n) \in \mathbb{R}^n$$

in this problem we'll write the list vertically

$$\begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}.$$

Now let $\mathbf{A}$ be an $m \times n$ matrix. Define $V = \{\, \mathbf{A}\mathbf{u}; \mathbf{u} \in \mathbb{R}^n \,\} \subset \mathbb{R}^m$. In words: V is the set of all possible vectors you can get by left-multiplying $\mathbf{A}$ onto some column vector $\mathbf{u}$. This set is called the "image" of the matrix $\mathbf{A}$ in the codomain $\mathbb{R}^m$.

1. Show that V is a subspace of $\mathbb{R}^m$.
2. Give an explicit parametrization for the subspace V in the cases

$$\text{(i)} \quad \mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \end{pmatrix}. \qquad \text{(ii)} \quad \mathbf{A} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \\ 3 & 1 \end{pmatrix}.$$

We'll describe two different approaches to this problem. The first will be an abstract set theoretic approach based on the definitions. This approach is the most elegant although some students will find it a bit abstract. The second approach will be a more concrete approach using the concept of span.

First solution to (1).

We'll check the three axioms in turn.

First axiom: Why does V contain the zero vector $\mathbf{0}_{m\times 1}$? Because this vector is the result of left-multiplying $\mathbf{A}$ onto the zero vector $\mathbf{0}_{n\times 1}$. That is:

$$\mathbf{0}_{m\times 1} = \mathbf{A}\mathbf{0}_{n\times 1}.$$

Second axiom: Why is V closed under vector addition? Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be two $m \times 1$ column vectors in V. According to the definition this means that there exist $n \times 1$ column vectors $\tilde{\mathbf{v}}_1$ and $\tilde{\mathbf{v}}_2$ such that

$$\mathbf{v}_1 = \mathbf{A}\tilde{\mathbf{v}}_1 \quad \text{and} \quad \mathbf{v}_2 = \mathbf{A}\tilde{\mathbf{v}}_2.$$

Now we check the sum:

$$\mathbf{v}_1 + \mathbf{v}_2 = \mathbf{A}\tilde{\mathbf{v}}_1 + \mathbf{A}\tilde{\mathbf{v}}_2 = \mathbf{A}\left(\tilde{\mathbf{v}}_1 + \tilde{\mathbf{v}}_2\right).$$

Thus $\mathbf{v}_1 + \mathbf{v}_2$ is also an element of V because it is the result of left-multiplying $\mathbf{A}$ onto some other column vector.

Third axiom: Why is V closed under scalar multiplication? Let $\mathbf{v}$ be some $m \times 1$ column vector in V. And let $r \in \mathbb{R}$. We must show that $r\mathbf{v}$ lies back in V. Well, according to the definition of V, if $\mathbf{v} \in V$ then there exists some $n \times 1$ column vector $\tilde{\mathbf{v}}$ such that $\mathbf{v} = \mathbf{A}\tilde{\mathbf{v}}$. Then:

$$r\mathbf{v} = r\mathbf{A}\tilde{\mathbf{v}} = \mathbf{A}\left(r\tilde{\mathbf{v}}\right).$$

Thus because $r\mathbf{v}$ is the result of multiplying some column vector by $\mathbf{A}$ we conclude that it lies back in V.

Second solution to (1).

Let $\mathbf{a_1}, \ldots \mathbf{a_n}$ denote the columns of $\mathbf{A}$ so that $\mathbf{A} = (\mathbf{a_1}, \ldots, \mathbf{a_n})$. Then

$$\begin{aligned} V &= \left\{ \mathbf{A} \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} \,\middle|\, \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} \in \mathbb{R}^n \right\} \\ &= \{ u_1\mathbf{a_1} + \cdots + u_n\mathbf{a_n}; u_1, \ldots, u_n \in \mathbb{R}\} \\ &= \operatorname{span}\{ \mathbf{a_1}, \ldots, \mathbf{a_n}\} \end{aligned}$$

hence V is a subspace. (The span of a collection of vectors is always a subspace by a theorem from the lectures.)

Solution to (2).

First consider the case

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

Then the set we want to study is

$$V = \left\{ \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} : u_1, u_2, u_3 \in \mathbb{R} \right\}$$

If we calculate the matrix product we rewrite this set

$$V = \left\{ \begin{bmatrix} u_1 + 2u_2 + 3u_3 \\ u_2 + u_3 \end{bmatrix} : u_1, u_2, u_3 \in \mathbb{R} \right\}.$$

This is a parametrization. We could rewrite this expression to get:

$$V = \left\{ u_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + u_2 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + u_3 \begin{bmatrix} 3 \\ 1 \end{bmatrix} : u_1, u_2, u_3 \in \mathbb{R} \right\}.$$

The other case is just as easy and we calculate

$$V = \left\{ \begin{bmatrix} u_1 \\ 2u_1 + u_2 \\ 3u_1 + u_2 \end{bmatrix} : u_1, u_2 \in \mathbb{R} \right\}.$$

Or we can write this

$$V = \left\{ u_1 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + u_2 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} : u_1, u_2 \in \mathbb{R} \right\}.$$

$\square$

Problem 9: proof of subspace

Consider two subspaces V_1 and V_2 of some $\mathbb{R}^n$. Show that if their union $V_1 \cup V_2$ is also a subspace of $\mathbb{R}^n$ then either $V_1 \subset V_2$, or $V_2 \subset V_1$, (or both, in which case the subspaces are equal).

Solution

We'll divide this into two cases. The first is the case $V_1 \subset V_2$. If this is true then we are done. So we focus on the case $V_1 \not\subset V_2$.

Assume $V_1 \not\subset V_2$. This means there is a vector $x \in V_1$ which does not lie in V_2, $x \notin V_2$. We will show that it must follow that all of $V_2 \subset V_1$. This will finish this case and establish the statement we are proving.

Take an arbitrary vector $y \in V_2$. We will explain why $y \in V_1$.

Because we are assuming that $V_1 \cup V_2$ is a subspace it is closed under vector addition, so we know that $x + y \in V_1 \cup V_2$. There are now two possibilities, at least one of which is true: (i) $x + y \in V_1$ (ii) $x + y \in V_2$.

Possibility (i): If $x + y \in V_1$ then $y = w - x$ for some $w \in V_1$. Thus $y \in V_1$.

Possibility (ii): If $x + y \in V_2$ then $x = w - y$ for some $w \in V_2$. But this is impossible because we chose x to be a vector from V_1 which doesn't lie in V_2. Thus this possibility doesn't happen.

So we conclude that Possibility (i) is the only possibility and every $y \in V_2$ lies in V_1.

□

Problem 10: proof of a subspace

Consider two subspaces V_1 and V_2 of some $\mathbb{R}^n$. Define the sum of these subspaces $V_1 + V_2$ to be the subset of $\mathbb{R}^n$ defined by the set theoretic expression

$$V_1 + V_2 = \{v_1 + v_2 | v_1 \in V_1, v_2 \in V_2\}.$$

In words: $V_1 + V_2$ is the set of all possible vectors in $\mathbb{R}^n$ you can get by taking a vector v_1 from V_1, a vector v_2 from V_2, and adding them together: $v_1 + v_2$.

1. Show that $V_1 + V_2$ is indeed a subspace of $\mathbb{R}^n$ containing both V_1 and V_2.
2. Moreover, explain why if W is *any* subspace of $\mathbb{R}^n$ containing both V_i then $V_1 + V_2 \subset W$.

Solution to 1.

To check this is a subspace we check the usual three properties.

Non-empty? Note that because V_1 and V_2 are both subspaces they both contain the zero vector. Thus $0 + 0 = 0$ lies in $V_1 + V_2$.

Closed under vector addition? Take two arbitrary elements of $V_1 + V_2$:

$$v_1 + v_2 \quad \text{and} \quad w_1 + w_2$$

where $v_1, w_1 \in V_1$ and $v_2, w_2 \in V_2$. Now check their sum

$$(v_1 + v_2) + (w_1 + w_2) = (v_1 + w_1) + (v_2 + w_2).$$

Because V_1 is a subspace, it is closed under vector addition, so $v_1 + w_1 \in V_1$. Similarly because V_2 is a subspace, it is closed under vector addition, so $v_2 + w_2 \in V_2$. Thus the sum of these two arbitrary vectors from $V_1 + V_2$ also lies in $V_1 + V_2$ because it is the sum of a vector from V_1 (namely $v_1 + w_1$) plus a vector from V_2 (namely $v_2 + w_2$).

Closed under scalar multiplication? Take an arbitrary real $r \in \mathbb{R}$ and an arbitrary vector from $V_1 + V_2$, denote it $v_1 + v_2 \in V_1 + V_2$. We must check

$$r(v_1 + v_2)$$

lies back in $V_1 + V_2$. We evaluate this expression:

$$r(v_1 + v_2) = rv_1 + rv_2.$$

Now because V_1 is a subspace, it is closed under scalar multiplication, so $rv_1 \in V_1$. Also because V_2 is a subspace, it is also closed under scalar multiplication, so $rv_2 \in V_2$. Thus this vector $rv_1 + rv_2$ lies back in $V_1 + V_2$ because it is the sum of a vector from V_1 and a vector from V_2.

That concludes the proof that $V_1 + V_2$ is a subspace.

Solution to 2.

Consider a subspace $W \subset \mathbb{R}^n$ containing both V_1 and V_2. Our task is to explain why

$$V_1 + V_2 \subset W.$$

An arbitrary vector in $V_1 + V_2$ is of the form $v_1 + v_2$ where $v_1 \in V_1$ and $v_2 \in V_2$.

Now because $V_1 \subset W$, we know that $v_1 \in W$.

Similarly, because $V_2 \subset W$, we know that $v_2 \in W$.

Thus, because W is a subspace, it is closed under vector addition, so the sum also

$$v_1 + v_2 \in W.$$

This shows that an arbitrary vector form $V_1 + V_2$ lies in W. Thus $V_1 + V_2 \subset W$.

$\square$

Comment: The way we capture this idea in words is: "V_1+V_2 is the smallest subspace containing both V_1 and V_2", because it lies in **any** subspace containing V_1 and V_2. In fact this provides an elegant alternative *definition* of V_1+V_2 as "the intersection of every subspace of $\mathbb{R}^n$ which contains both V_1 and V_2".

Comment: If you are still curious after this problem: Once you've understood sums of subspaces, you might want to google "direct sum of subspaces" to learn about it direct sums, which are an important special case of the sum construction we just studied. They are denoted by the special notation $V_1 \oplus V_2$.

Problem 11: proof of a subspace (from 2020 midterm.)

Two vectors in $\mathbb{R}^n$

$\vec{v} = (v_1, \ldots, v_n)$

$\vec{w} = (w_1, \ldots, w_n)$

are said to be **orthogonal** if their scalar product, which is defined by the expression

$$\vec{v} \cdot \vec{w} = v_1 w_1 + v_2 w_2 + \ldots + v_n w_n \ ,$$

equals zero.

Now fix a non-zero vector $\vec{v} \in \mathbb{R}^n$.

(a) Show directly that the three axioms of a subspace hold for the set $S \subset \mathbb{R}^n$ of all vectors in $\mathbb{R}^n$ orthogonal to $\vec{v}$. In set theory notation we would write

$$S = \{\vec{w} \in \mathbb{R}^n ; \vec{v} \cdot \vec{w} = 0\} \subset \mathbb{R}^n .$$

(That is: do not use the homogeneous system theorem, but use the axioms directly in this problem.)

(b) Show that every vector $\vec{w} \in \mathbb{R}^n$ can be expressed in the form

$$\vec{w} = c\vec{v} + \vec{z}$$

where $c \in \mathbb{R}$ and $\vec{z} \in S$.

Solutions to (a).

(*1st axiom: S is non-empty.*)

S contains the zero vector $\vec{0} = (0, \ldots, 0)$ because

$$\vec{v} \cdot \vec{0} = v_1 * 0 + \ldots + v_n * 0 = 0.$$

(*2nd axiom: S is closed under vector addition.*)

Assume that $\vec{x} = (x_1, \ldots, x_n)$ and $\vec{y} = (y_1, \ldots, y_n)$ are arbitrary vectors lying in S. This means that

$$v_1 x_1 + \ldots + v_n x_n = 0 \ \text{ and } \ v_1 y_1 + \ldots + v_n y_n = 0.$$

Now we test whether $\vec{x} + \vec{y} = (x_1 + y_1, \ldots, x_n + y_n)$ lies in S. We need

$$\vec{v} \cdot (\vec{x} + \vec{y}) = 0.$$

The left-hand-side equals:

$$\vec{v} \cdot (\vec{x} + \vec{y}) = v_1(x_1 + y_1) + \ldots + v_n(x_n + y_n).$$

This equals:

$$(v_1 x_1 + \ldots + v_n x_n) + (v_1 y_+ \ldots + v_n y_n).$$

This equals

$$0 + 0 = 0.$$

Thus S is closed under vector addition.

(*3rd axiom: S is closed under scalar multiplication.*)

Assume that $\vec{x} = (x_1, \dots, x_n)$ is an arbitrary vector lying in S and $c \in \mathbb{R}$. This means that

$$v_1x_1 + \ldots + v_nx_n = 0.$$

Now we test whether $c\vec{x} = (cx_1, cx_2, \dots, cx_n)$ lies in S. We need that

$$\vec{v} \cdot (c\vec{x}) = 0.$$

The left-hand-side equals

$$v_1(cx_1) + \ldots + v_n(cx_n).$$

This equals

$$c\,(v_1x_1 + \ldots + v_nx_n) = c * 0 = 0.$$

Thus $c\vec{x} \in S$ and S is closed under scalar multiplication.

Solutions to (b).

We are given $\vec{v} = (v_1, \dots, v_n)$ and $\vec{w} = (w_1, \dots, w_n)$.
It is enough for us to find a real number c such that

$$\vec{v} \cdot (\vec{w} - c\vec{v}) = 0 \quad (\star).$$

because then we just write

$$\vec{w} = c\vec{v} + (\vec{w} - c\vec{v})$$

to find the required decomposition into a vector in the span of $\vec{v}$ and a vector in S.

Using the given formula for inner product, the left-hand-side of $(\star)$ equals

$$v_1(w_1 - cv_1) + \ldots + v_n(w_n - cv_n).$$

This equals:

$$(v_1w_1 + \ldots + v_nw_n) - c(v_1^2 + \ldots + v_n^2).$$

Note that because $\vec{v} \neq \vec{0}$ the expression

$$v_1^2 + \ldots + v_n^2 \neq 0.$$

Thus equation $(\star)$ can be simply solved:

$$c = \frac{v_1w_1 + \ldots + v_nw_n}{v_1^2 + \ldots + v_n^2}.$$

$\square$

Comments

This was actually an elementary algebraic problem but formulated in an unfamiliar setting. You didn't need any special prior knowledge about inner product or geometry to solve this problem, just the ability to see through the unfamiliar language to the simple algebraic problem you were given. This is also something students need to learn at university: being able to bring what they know to an unfamiliar situation.